Identifying Farmers at Risk of Crop Failure

Sankalp Dwivedi IPM06141

IT Applications

Dr. Karan Verma

March 3rd, 2026

Integrated Programme in Management

IIM Rohtak

Introduction

In many countries, especially developing economies like India, a significant portion of the working population relies on farmingdirectly or indirectly. The agricultural industry, while crucial, is inherently uncertain and risky. There are many factors farmers need to consider when deciding what type of crop to grow, when to plant it, and how many resources to commit to growing it, with very little information and significant environmental unpredictability. The unpredictability of factors such as rainfall amounts, soil quality, temperature, and seasonal trends can severely impair agricultural productivity, leading to crop failure.

Crop failure not only results in financial losses for farmers but can also affect food supply chains and regional economic stability. Traditionally, India rural farmers have made decisions based on experience, historical practices, and local knowledge passed down through generations. While those methods do work, they may not account for the complex interactions among environmental variables, crop type, and farm practices. As a result, there is increasing interest in using data-driven approaches to support agricultural decision-making.

This project tries to apply machine learning techniques to agricultural production data to better understand patterns of crop productivity and assess the risk of crop failure. Through a hybrid approach, the project aims to analyze farming characteristics such as crop type, cultivated area, season, and regional distribution to identify productivity patterns and predict risk levels.

Dataset and Project Overview

The first dataset used in the project is a crop production dataset that contains historical information on agricultural output across different regions. It provides information on where crops have been produced throughout history, on a per-state and district basis (aggregated data); therefore, we can examine all the places where a given crop has been grown to build a picture of where the product is grown and the amount produced in these areas. This dataset includes the number of acres planted for each crop, the crop types, and the amount produced from each planting. From this data, an additional measure, “yield”, was calculated by dividing production by the cultivated area. Yield represents the productivity of land and provides a useful way to compare crop performance across different farms and regions.

The second dataset used in the project is a crop-recommendation dataset containing environmental and soil-related variables. This dataset includes information about soil nutrients such as nitrogen, phosphorus, and potassium levels, as well as climatic conditions such as temperature, humidity, rainfall, and soil pH. Each observation in this dataset is associated with a crop label indicating which crop is best suited to the given environmental conditions. This dataset allows the project to explore how environmental variables influence crop suitability.

Methodology

The first step of the analysis involved data preprocessing. To ensure reliable analysis, the data were cleaned by removing incomplete entries and observations with zero or missing cultivated area or production values.

Using the yield values, farms were categorized into three groups to represent different levels of productivity risk. Farms with the lowest 25 percent of yield values were labeled as high risk, farms within the middle range were labeled as medium risk, and farms with the highest yield values were labeled as low risk. This classification created the target variable used for predictive modeling.

The project also applied unsupervised learning techniques to identify hidden patterns in the data. The K-Means clustering algorithm was used to group farms based on similarities in area, production, and yield. Clustering identifies natural groupings within a dataset without predefined labels. The elbow method was used to determine the number of clusters. This method assesses how clustering performance changes with increasing cluster count. The analysis suggested that three clusters provided a meaningful balance between simplicity and explanatory power. These clusters represent farms with different productivity characteristics.

In addition to clustering, the project implemented supervised machine learning models to predict farm risk levels. Three models were employed in this analysis. The first model was Logistic Regression, which estimates the probability that a farm belongs to a particular risk category based on input features such as crop type, season, location, and cultivated area.

The second model was a Random Forest classifier, which is a more advanced ensemble learning method. Random Forest builds multiple decision trees and combines their predictions through

voting or averaging to improve accuracy and reduce overfitting. This model is well-suited for complex datasets where relationships between variables may not be strictly linear.

Finally, a hybrid model was developed by using the K-Means cluster label as an additional feature in the Random Forest model. By combining clustering with predictive modeling, the hybrid approach aims to improve the model's ability to detect farms that may be at higher risk of crop failure.

The performance of the models was evaluated using metrics such as accuracy, precision, recall, and F1-score, which provide a comprehensive assessment of how well the models classify farms into different risk categories.

Output and Interpretation

The crop suitability model is one of the major outputs of this analysis, created using environmental factors (e.g., humidity, temperature, rainfall, and soil nutrients) to predict which crops grow best under each condition. The confusion matrix showed that most crops in the dataset were correctly predicted, supporting the hypothesis that soil and climate have a significant impact on whether a crop will thrive in a given environment.

Several visualizations were generated to explore the structure of the crop production dataset. The state-level production chart shows how output varies significantly across regions. Certain states, like Kerala, contribute a much larger share to the overall agricultural production. However, it is important to note that such a difference could be due to data-collection bias.

Perhaps the most important visualization was the crop distribution heatmap by state, which reveals patterns of regional crop specialization. The heatmap shows that some crops are strongly associated with particular regions. For example, some crops appear much more frequently in certain states due to favorable climatic conditions, soil characteristics, and established farming practices. This confirms that agricultural production is heavily influenced by local environmental conditions and that farmers can use these factors to select crops.

The seasonal crop distribution heatmap further highlights how farming activity changes across different agricultural seasons. Certain crops dominate particular seasons, reflecting traditional

agricultural cycles. For instance, crops commonly associated with monsoon conditions appear more frequently in one season, while others that require cooler conditions appear in another. These patterns illustrate the importance of seasonal timing in crop cultivation.

The K-Means clustering analysis provided additional insight into farm productivity patterns. Farms were grouped into three clusters based on similarities in cultivated area, production, and yield. The cluster visualization shows that farms with similar productivity characteristics tend to form distinct groups. The summary table for clusters showed differences in average area, production, and yield, indicating that farms can be grouped by productivity levels.

Practical Insights

The analysis of the crop data indicates a number of significant trends among the farms used for agriculture, which are illustrated graphically in the data. The heat map of crop distribution makes it clear that nearly every crop type has been assigned to specific states as its “home.” This demonstrates that regional influences affect all crops, including climate, soils, irrigation availability, and traditional farming practices.

Another key insight comes from the seasonal crop distribution patterns. The analysis shows that significant differences exist in the types of crops selected for each agricultural season. Those differences reflect the impact of seasonal weather conditions or local environments on each specific season, including rainfall patterns and temperature cycles. Crops that require a lot of water are usually grown during the monsoon season, while crops that require cooler temperatures or a lack of rainfall are grown during the other seasons.

The clustering analysis also provided valuable insights into farm productivity. The clustering algorithm grouped farm plots based on land area, crop production level, and crop yield. The predictive models further showed that it is possible to estimate farm risk using features such as crop type, state, season, and cultivated area. The findings of this project confirm that agricultural productivity is not completely random and that patterns are present within the agricultural datasets analyzed. More specifically, hybrid models are shown to achieve marginally improved performance compared to Random Forest models, suggesting that combining unsupervised

(descriptive) learning with predictive modeling provides an opportunity to detect more patterns in agricultural data.

The project illustrates that practical applications of machine learning techniques can be used in real-world agricultural settings. For instance, combining historical production data with environmental data to build predictive models can identify farms/regional areas with a higher likelihood of producing low yields/experiencing low production rates. From this information, farmers, agricultural planners, and policymakers can make better-informed decisions. Systems based on similar models could be used to recommend crops best suited to specific environmental conditions.

Conclusion

This project demonstrated how machine learning algorithms can be applied to agricultural production data to understand crop productivity patterns and assess the likelihood of crop failure. By analysing exploratory data, applying different clustering methods, and using predictive modelling, we were able to identify crop production patterns within a given geographic area, determine seasonal trends in farming, and identify differences in productivity across individual farms.

The results show that machine learning models can successfully identify meaningful patterns in agricultural datasets and provide reasonably accurate predictions of crop productivity risk. Visualizations and clustering analysis further helped reveal structural differences between farms and regional agricultural practices. In summary, this project demonstrates how data analytics and machine learning can deepen our understanding of farming systems and how data-driven technologies can enable better decision-making in the agricultural industry.